

Group 9 Review 1

Product Name

SentraGrade

Slogan

"Not just sorting seeing what doesn't belong, and why."

1. Introduction

Automated grading lines in the fruit and vegetable export and packaging industry rely on machine vision to sort produce by size, ripeness, and surface quality at speeds no human inspector could match. These systems are trained as closed-set classifiers they recognize a fixed catalogue of known produce types and known defect categories. The gap this creates is structural: when something outside that catalogue reaches the line a foreign object, packaging debris, or a produce variety the system was never trained on a closed-set classifier is mathematically forced to assign it to one of its known categories anyway, since a softmax output must sum to one regardless of whether the input actually belongs. SentraGrade addresses this by reframing grading-line vision as an explainable out-of-distribution (OOD) detection problem: reject what doesn't belong, and show visually why.

2. Problem Statement

Automated fruit and vegetable grading systems rely on vision models trained to recognize a fixed set of known produce categories. In real-world deployment, these systems inevitably encounter foreign objects and unrecognized items never part of the training distribution packaging debris, contaminants, or produce varieties the model has never seen which a standard closed-set classifier will confidently but incorrectly assign to a known class rather than flag as anomalous. This project proposes an explainable out-of-distribution (OOD) detection framework for automated grading systems, designed to reject such unrecognized inputs rather than misclassify them, while providing visual justification for every accept/reject decision.

3. Motivation

- **Food safety and export compliance:** undetected foreign material in packed produce is a direct contamination and liability risk, and export markets increasingly demand traceable, auditable quality control rather than opaque pass/fail decisions.

- **Cost of manual inspection:** grading lines that still rely on human oversight for edge cases (anything outside the trained defect catalogue) reintroduce the labor cost and inconsistency automation was meant to eliminate.
- **Trust and auditability gap in current commercial systems:** as the related-products review below shows, deployed industrial sorters are highly capable at detecting known, pre-trained defect types, but function as black boxes an operator sees a reject decision, not a visual reason for it. This is a real limitation when a batch is disputed or a regulator asks why product was rejected.
- **Closed-set blind spot:** existing systems are only as good as their trained defect catalogue; a genuinely novel foreign object type is a blind spot by design, not an edge case they were built to handle.

4. Research Questions / Gaps to be Addressed

- **RQ1:** Can a leave-one-class-out evaluation protocol, using a held-out known produce class as a stand-in for "unrecognized object," reliably approximate real-world foreign-object rejection when no annotated contaminant dataset exists?
- **RQ2:** Does the quality of learned representations (supervised vs. self-supervised pretraining; CNN vs. Vision Transformer backbones) affect OOD separability more than the choice of OOD scoring function itself?
- **RQ3:** Can Grad-CAM / attention-based explainability outputs double as weakly-supervised localization of the anomalous region, without manually annotated defect masks?
- **Gap addressed:** Commercial grading-line AI is optimized for closed-set defect classification accuracy and throughput, not for explainability of rejection decisions or for detecting genuinely novel object classes outside the trained catalogue this project targets that specific combination.

5. Related Products and Their Details

TOMRA 5B / 5C Sorting Machines URL: <https://www.tomra.com/food/machines> Features: Detects foreign materials like stem stalks and Datura while sorting by shape, size, and length-to-width ratio, with 360° inspection using surround-view technology powered by camera and laser sensors. Combines high-performance lasers, BSI+ sensors, and machine learning analysis to identify every piece of product and all foreign material. Limitations: The system is trained on and detects a defined library of known foreign material and defect types (e.g., stem stalks, specific toxins); it is a closed-set defect detector, and the decision output is a sort/reject action rather than a visual, human-interpretable explanation of *why* an item was rejected.

Compac Spectrim Optical Sorting Platform URL: <https://www.compacsort.com> (product coverage via <https://content.compacsort.com/blog/spectrim-new-optical-sorting-platform>) Features: Uses machine-learning algorithms to automate defect detection and classification tasks operators would otherwise perform manually, capturing up to 500 images of each piece of

produce using multiple wavelengths, each configurable to target specific defects. Processes up to 12 pieces of fruit per second while detecting hard-to-find external blemishes. Limitations: Like TOMRA's systems, Spectrim's machine-learning defect classification is trained against a known defect taxonomy; genuinely novel object types outside that taxonomy are not addressed as a distinct detection category, and no explainability layer is exposed to justify individual rejection decisions to an operator or auditor.

Common gap across both: neither system publishes an explicit "unknown/novel object" detection mode distinct from known-defect classification, and neither exposes a visual explanation (e.g., heatmap) for individual reject decisions both are the specific gaps SentraGrade is designed to fill.

6. SDG Mapping

6.1 SDG Identified

- **SDG 12: Responsible Consumption and Production** (primary)
- **SDG 2: Zero Hunger** (secondary)

6.2 How the Case Study Relates to the SDG Grading errors both false rejections of good produce and false acceptances of contaminated batches directly contribute to food loss and waste, which SDG 12 explicitly targets in its food-loss-reduction indicators. A more reliable, explainable rejection system reduces both failure modes: it catches genuine foreign objects/contamination more reliably than a forced-choice closed-set classifier, while its explainability layer allows operators to catch and correct false rejections of good produce, directly reducing avoidable food waste. This also connects to SDG 2, since reduced post-harvest loss in export-oriented grading lines increases the effective food supply reaching consumers from the same volume of harvested produce.

7. Dataset Details

Note on hosting: this dataset is hosted on **Figshare**, not IEEE Dataport
https://figshare.com/articles/dataset/TriModal_Ripeness_6/30783827

7.1 Dataset Description TR-6 (TriModal Ripeness 6) is a dataset comprising approximately 29,000 images (combined sRGB and IR-Fusion thermal) totaling 18.99 GB, collected across six produce classes guava, carrot, tomato, Indian gooseberry, banana, and mango captured over each specimen's decomposition period to study visual, thermal, and chemical spoilage indicators together. This project uses the sRGB subset (~10,000 images) as its working dataset; IR-Fusion thermal imagery and methane sensor readings are part of the source dataset but outside the current project's scope.

7.2 Specific Challenges Addressed in the Dataset

- **Class imbalance:** per-category image counts vary substantially (e.g., Tomato ~4,488 vs. Banana ~646 sRGB images), requiring class-weighted loss or resampling during training.
- **Same-specimen leakage:** each fruit was photographed repeatedly across its decomposition period, so images of the same physical specimen must be grouped and split by specimen/session ID rather than by individual image, to prevent train/test leakage.
- **Device and resolution inconsistency:** images were captured across three different phone models with differing resolutions and sensors, requiring standardized resizing and a check that device usage doesn't correlate with class label.
- **Preprocessing pipeline divergence:** supervised and ViT backbones use standard ImageNet normalization, while the self-supervised (SimCLR/DINOv2) branch requires its own augmentation pipeline (random crop, color jitter, blur), which must be handled distinctly per backbone branch in the pipeline.

8. Novelty in the Product

- Reframes grading-line defect detection as an **explainable open-set problem**, rather than the closed-set classification approach used by deployed commercial systems like TOMRA and Compac.
- Introduces a **leave-one-class-out evaluation protocol** as a defensible proxy for genuine foreign-object rejection, in the absence of any publicly available annotated contaminant dataset for produce grading.
- Runs a **dual ablation study** representation quality (supervised vs. self-supervised vs. ViT) crossed with OOD scoring method (energy score, MSP, prototype distance) producing a comparative analysis of what actually drives detection reliability, rather than reporting a single arbitrary configuration.
- Uses **Grad-CAM / attention maps not just for post-hoc justification but as weak, annotation-free localization** of the anomalous region, closing the explainability gap identified in both commercial systems reviewed above, where a reject decision is delivered with no visual justification.